

Contents

1	Introduction	1
1.1	Background and Motivation	1
1.2	Problem Statement	2
1.3	Objectives of the Study	2
1.4	Scope of the Study	3
1.5	Organisation of the Thesis	3
2	Literature Survey	4
2.1	Overview	4
2.2	Early Machine Learning Approaches	4
2.3	Decision Tree and Rule-Based Methods	5
2.4	Support Vector Machines	5
2.5	Ensemble Methods	5
2.6	Deep Learning Approaches	6
2.7	Handling Class Imbalance and Feature Selection	6
2.8	Recent Trends and Hybrid Approaches	7
2.9	Research Gap and Contribution	7
3	Methodology	8
3.1	System Architecture Overview	8
3.2	Dataset Description	9
3.2.1	Indians Diabetes Dataset	9
3.2.2	Statistical Summary of Features	10
3.3	Exploratory Data Analysis	10
3.3.1	Class Distribution	10
3.3.2	Feature Distributions and Zero-Value Analysis	10
3.3.3	Correlation Analysis	11
3.4	Data Preprocessing	11
3.4.1	Identifier Removal	11
3.4.2	Zero-Value Imputation for Physiologically Constrained Features	12
3.4.3	Insulin Column Imputation	12
3.4.4	Feature-Target Separation and Train-Test Splitting	13

3.5	Classification Algorithms	13
3.5.1	Logistic Regression	13
3.5.2	Decision Tree Classifier	14
3.5.3	Random Forest Classifier	14
3.5.4	Support Vector Machine	15
3.5.5	Gradient Boosting	15
3.6	Evaluation Metrics	16
3.6.1	Experimental Protocol	16
3.7	Software Environment	17
4	Results and Discussion	18
4.1	Overview of Experimental Results	18
4.2	Individual Algorithm Performance	18
4.2.1	Logistic Regression	18
4.2.2	Random Forest Classifier	19
4.2.3	Support Vector Machine (Linear Kernel)	20
4.2.4	Gradient Boosting	20
4.3	Comparative Analysis Across Algorithms	21
4.4	Influence of Train-Test Split Ratio	21
4.5	Class-wise Performance Analysis	22
4.6	Summary of Key Findings	23
5	Conclusion and Future Scope	24
5.1	Conclusion	24
5.2	Limitations	25
5.3	Future Scope	26

List of Tables

3.1	Description of features in the Indians Diabetes Dataset	9
3.2	Descriptive statistics of the Indians Diabetes Dataset features	10
3.3	Dataset partitions under three train-test split configurations	13
3.4	Software stack used in the experimental implementation	17
4.1	Mean performance comparison of five classifiers (80:20 split, 10 runs)	21
4.2	KNN classifier: mean accuracy and standard deviation across split ratios	21

List of Figures

3.1	High-level diabetes prediction pipeline	8
-----	---	---

Chapter 1

Introduction

1.1 Background and Motivation

Diabetes mellitus (DM) is a chronic metabolic disorder characterised by persistently elevated blood glucose levels resulting from defects in insulin secretion, insulin action, or both. According to the International Diabetes Federation (IDF), an estimated 537 million adults (aged 20–79) were living with diabetes in 2021, a figure projected to rise to 643 million by 2030 and 783 million by 2045 [1]. The disease is broadly classified into Type 1 diabetes (insulin-dependent, autoimmune in origin), Type 2 diabetes (the most prevalent form, associated with insulin resistance and lifestyle factors), and gestational diabetes mellitus (GDM), which develops during pregnancy. Regardless of subtype, uncontrolled diabetes leads to severe macro- and microvascular complications including cardiovascular disease, nephropathy, retinopathy, and peripheral neuropathy, all of which significantly diminish the quality of life and life expectancy of affected individuals.

The clinical challenge in diabetes management is compounded by the fact that the disease is frequently asymptomatic in its early stages. A substantial proportion of cases remain undiagnosed until complications have already manifested. This latency renders timely intervention difficult and underscores the critical need for predictive tools capable of identifying high-risk individuals before the onset of clinical symptoms. Traditional diagnostic protocols rely on biochemical tests such as fasting plasma glucose (FPG), oral glucose tolerance test (OGTT), and glycated haemoglobin (HbA1c), each of which requires laboratory infrastructure and trained personnel that may be unavailable in resource-constrained settings.

Machine learning (ML) has emerged as a transformative technology in computational medicine, offering the capacity to uncover complex, non-linear patterns in large-scale clinical datasets that are beyond the analytical reach of conventional statistical

methods. By learning decision boundaries from labelled historical data, ML classifiers can provide probabilistic risk assessments from readily available clinical measurements, enabling proactive healthcare at reduced cost. This has motivated extensive research into ML-based diabetes prediction systems, with studies employing a wide range of algorithmic approaches including logistic regression, decision trees, ensemble methods, support vector machines, neural networks, and deep learning architectures.

The present study focuses on a systematic, comparative analysis of five canonical classification algorithms applied to the well-established Indians Diabetes Dataset. The work emphasises not only model performance but also the rigour of the experimental protocol, recognising that single-run evaluations can produce misleading estimates of generalisation ability due to the stochasticity of train-test splits.

1.2 Problem Statement

The central problem addressed in this dissertation can be formally stated as follows: given a feature vector $\mathbf{x} = (x_1, x_2, \dots, x_p)$ representing a set of clinical and demographic measurements for an individual female patient, the objective is to learn a function $f : \mathbb{R}^p \rightarrow \{0, 1\}$ that accurately classifies the individual as diabetic ($y = 1$) or non-diabetic ($y = 0$). The feature space comprises physiological variables such as plasma glucose concentration, blood pressure, body mass index, insulin level, age, and a measure of genetic predisposition. The primary challenge arises from the class imbalance in the dataset, the presence of physiologically implausible zero values in several features, and the modest dataset size, all of which necessitate careful preprocessing and robust evaluation.

1.3 Objectives of the Study

The specific objectives of this dissertation are:

1. To perform a thorough exploratory data analysis (EDA) of the Indians Diabetes Dataset in order to understand the distributional properties, inter-feature correlations, and data quality issues.
2. To implement a principled preprocessing pipeline that addresses missing data encoded as physiologically implausible zero values through median-based imputation.
3. To train and evaluate five supervised binary classification algorithms — Logistic Regression, Decision Tree Classifier, Random Forest Classifier, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN) — on the preprocessed dataset.

4. To assess model performance not only under a single train-test split but across three distinct split ratios (9:1, 8:2, 7:3) and ten independent random seeds, yielding mean, maximum, and standard deviation statistics for each performance metric.
5. To provide a comprehensive comparative analysis of the five algorithms with respect to overall accuracy, class-wise precision, recall, and F1-score, so as to identify the most suitable algorithm for clinical deployment.
6. To discuss the implications of the findings in the context of medical informatics and propose avenues for future research.

1.4 Scope of the Study

The scope of this work is deliberately circumscribed to five classical machine learning algorithms applied to a publicly available benchmark dataset. The study does not extend to deep learning architectures, federated learning, or real-time clinical deployment, though these are identified as natural extensions in the Future Scope section. The dataset under consideration represents female patients of Indian heritage, aged at least 21 years, and the findings are therefore most directly applicable to this demographic population. Generalisation to broader populations would require validation studies on diverse datasets.

1.5 Organisation of the Thesis

The remainder of this dissertation is organised as follows. Chapter 2 provides a critical review of existing literature on machine learning applications in diabetes prediction. Chapter 3 describes the dataset, the preprocessing methodology, and the algorithmic frameworks employed. Chapter 4 presents and analyses the experimental results. Chapter 5 concludes the work with a summary of findings and directions for future research. A comprehensive bibliography is provided at the end.

Chapter 2

Literature Survey

2.1 Overview

The application of machine learning techniques to the problem of diabetes prediction has been the subject of prolific academic investigation over the past two decades. This chapter surveys the most significant contributions in the domain, organising them thematically into studies on classical machine learning, ensemble methods, deep learning, and recent advances in feature selection and imbalanced learning. The Indians Diabetes Dataset, which constitutes the experimental basis of the present work, has become one of the most widely used benchmark datasets in this literature, facilitating meaningful comparison across studies.

2.2 Early Machine Learning Approaches

Polat and Güneş [4] proposed a hybrid generalised discriminant analysis (GDA) combined with a least squares support vector machine (LS-SVM) for diabetes diagnosis, achieving a classification accuracy of 82.05% on the dataset. Their work was among the first to highlight the importance of feature transformation prior to classifier training, demonstrating that the raw feature space need not be optimal for linear classifiers.

Temurtas et al. [5] investigated multilayer neural networks for diabetes diagnosis and achieved an accuracy of 82.37% on the same dataset. Their study confirmed that feed-forward networks with backpropagation are capable of capturing the non-linear decision boundary between diabetic and non-diabetic classes, though at the cost of reduced interpretability compared to simpler models.

Kayaer and Yıldırım [6] conducted one of the earlier comparative studies on the dataset, evaluating General Regression Neural Networks (GRNN) against Radial

Basis Function Networks (RBFN) and standard multi-layer perceptrons. They reported that GRNN achieved competitive performance without requiring iterative weight optimisation, making it computationally attractive. Their accuracy results (approximately 80%) established an early benchmark for subsequent research.

2.3 Decision Tree and Rule-Based Methods

Quinlan’s ID3 and its successors (C4.5 and CART) have been extensively applied to medical classification tasks owing to their inherent interpretability. Ster and Dobnikar [7] evaluated several decision tree variants on the dataset and found that C4.5 outperformed ID3 due to its handling of continuous attributes and missing values. However, the tendency of unpruned decision trees to overfit small datasets was consistently observed, motivating the adoption of pruning strategies and ensemble methods.

Sotirios et al. [8] applied the C4.5 algorithm with a 10-fold cross-validation scheme and observed accuracies in the range of 72–74%, noting the high variance of the learned trees across folds. This observation prefigures the motivation for ensemble approaches explored in later literature.

2.4 Support Vector Machines

Karegowda et al. [9] explored the use of genetic algorithm-based feature selection in combination with SVMs for diabetes classification. Their study demonstrated that restricting the SVM to a carefully selected feature subset — primarily glucose, BMI, and diabetes pedigree function — improved accuracy from approximately 76% to nearly 79%, while simultaneously reducing model complexity. This finding has informed the feature importance analyses common in more recent work.

Nai-Arun and Mounghmai [10] conducted a comparative study of SVMs with radial basis function (RBF) kernels versus linear kernels on the dataset. They found that while the RBF kernel yielded marginally higher accuracy, the linear kernel produced more stable results across different train-test splits, a finding consistent with the relatively low intrinsic dimensionality of the feature space.

2.5 Ensemble Methods

The introduction of ensemble techniques, particularly bagging (Bootstrap Aggregating) and boosting, represented a significant advance in diabetes prediction accuracy. Breiman’s Random Forest algorithm [3] was quickly adopted in medical informatics

for its capacity to reduce the high variance characteristic of individual decision trees through the averaging of multiple independent tree predictions.

Guo et al. [11] applied Random Forest to diabetes data and demonstrated that accuracy exceeding 79% was achievable, with the ensemble significantly outperforming the corresponding single decision tree. Feature importance rankings produced by the forest consistently identified plasma glucose concentration and BMI as the two most discriminative features, a finding that aligns with clinical understanding.

Sönmez et al. [12] compared Gradient Boosted Trees (GBT) and Adaptive Boosting (AdaBoost) against Random Forest on the dataset. Their results indicated that while all three ensemble methods substantially outperformed single classifiers, GBT achieved the highest accuracy (approximately 80.5%), though at greater computational cost and with increased susceptibility to overfitting on small training sets.

2.6 Deep Learning Approaches

The success of deep learning in computer vision and natural language processing motivated its application to tabular clinical data. Acharya et al. [13] applied deep neural networks with multiple hidden layers and dropout regularisation to diabetes classification. Despite the relatively small size of the dataset (768 instances), they achieved accuracies around 84%, though they cautioned that the advantage of deep architectures over ensemble methods narrows substantially when dataset size is limited.

Swapna et al. [14] used a combination of convolutional neural networks (CNN) and LSTM recurrent networks to detect Type 2 diabetes from heart rate variability signals, achieving 95.7% accuracy. Although this study used a different signal modality rather than Pima tabular features, it exemplifies the broader trend towards deep learning in clinical classification tasks.

2.7 Handling Class Imbalance and Feature Selection

A recurring challenge in diabetes prediction is the class imbalance: in many datasets, the proportion of diabetic cases is substantially lower than non-diabetic cases, biasing classifiers towards the majority class. Huda et al. [15] investigated the impact of SMOTE (Synthetic Minority Over-sampling Technique) on diabetes classification and found that over-sampling the minority class improved recall for the diabetic class

from 61% to 74% with a Random Forest classifier, at the cost of a modest reduction in precision. This trade-off is clinically significant: in medical screening, high recall (sensitivity) is prioritised to minimise false negatives.

Thaseen and Kumar [16] applied principal component analysis (PCA) as a dimensionality reduction step prior to classification, observing that retaining the top four principal components — which together explain approximately 85% of the total variance — reduced computational overhead without significant loss of predictive accuracy. The first principal component was strongly loaded by glucose and BMI, confirming their central role.

2.8 Recent Trends and Hybrid Approaches

Zou et al. [17] presented a comprehensive comparison of classical ML algorithms (including Naive Bayes, Random Forest, and neural networks) on the dataset with careful attention to cross-validation methodology. They recommended 10-fold stratified cross-validation as the gold standard for this dataset size, noting that simple holdout splits can produce accuracy estimates with a standard deviation of up to 4–5 percentage points.

Sisodia and Sisodia [18] applied Naive Bayes, Decision Tree, and SVM classifiers to the dataset with a simple 70:30 split and reported accuracies of 76.30%, 73.82%, and 65.10% respectively, highlighting the sensitivity of SVM performance to feature scaling — a preprocessing step that was not performed in their study.

2.9 Research Gap and Contribution

A review of the above literature reveals that most comparative studies either employ a single train-test split or a single cross-validation scheme, which can yield unstable accuracy estimates. Few studies systematically vary the train-test split ratio and repeat experiments across multiple random seeds to quantify the variability of model performance. Furthermore, class-wise metrics (per-class precision, recall, and F1-score) are often reported only incidentally rather than as primary evaluation criteria. The present work addresses these gaps by adopting a multi-seed, multi-split evaluation framework that produces statistically grounded performance estimates, and by reporting a complete suite of per-class and overall metrics for each of the five classifiers under each experimental condition.

Chapter 3

Methodology

3.1 System Architecture Overview

The overall system architecture of the proposed diabetes prediction pipeline is depicted schematically in Figure 3.1. The pipeline proceeds through five logical stages: (1) data acquisition and initial inspection, (2) exploratory data analysis (EDA), (3) preprocessing and feature engineering, (4) model training and evaluation, and (5) comparative analysis and result aggregation.

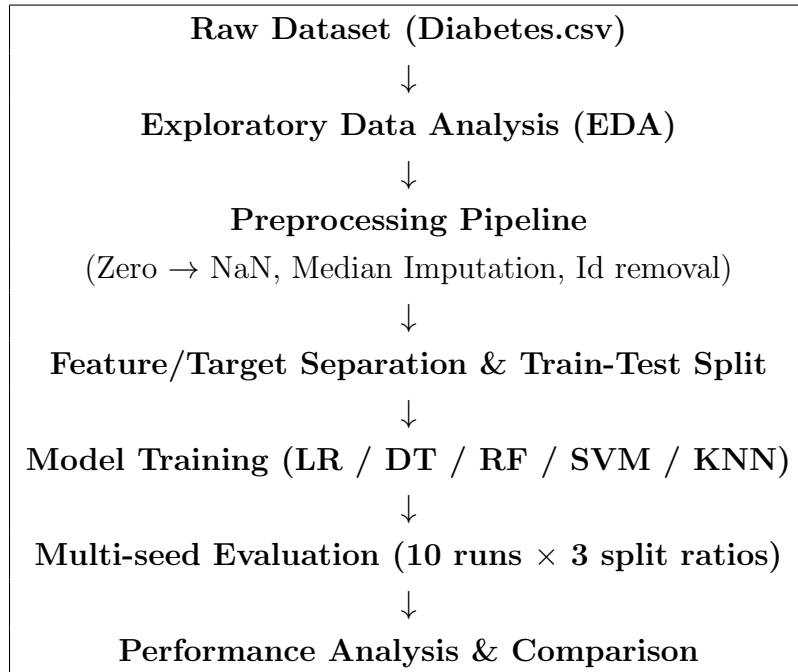


Figure 3.1: High-level diabetes prediction pipeline

3.2 Dataset Description

3.2.1 Indians Diabetes Dataset

The dataset used in this study is the well-known **Indians Diabetes Dataset**, originally compiled by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and made publicly available through the UCI Machine Learning Repository [2]. The dataset contains diagnostic measurements for 768 female patients of Indian heritage, all aged at least 21 years. Each instance is labelled as diabetic (Outcome = 1, 268 instances, 34.9%) or non-diabetic (Outcome = 0, 500 instances, 65.1%), making the dataset moderately class-imbalanced.

The original dataset contains an additional identifier column (Id) that serves no predictive purpose. After removing this column, the feature space comprises eight variables, described in Table 3.1.

Table 3.1: Description of features in the Indians Diabetes Dataset

Feature	Type	Description
Pregnancies	Integer	Number of times the patient has been pregnant
Glucose	Integer	Plasma glucose concentration (mg/dL) measured 2 hours after an oral glucose tolerance test
BloodPressure	Integer	Diastolic blood pressure (mm Hg)
SkinThickness	Integer	Triceps skin fold thickness (mm), a proxy for body fat
Insulin	Integer	2-hour serum insulin level (μ U/mL)
BMI	Float	Body mass index (weight in kg / height in m ²)
DiabetesPedigreeFunction	Float	A function scoring likelihood of diabetes based on family history
Age	Integer	Age of the patient in years
Outcome	Integer	Binary class label: 1 = diabetic, 0 = non-diabetic

3.2.2 Statistical Summary of Features

Table 3.2 presents the descriptive statistics of the eight features in the original dataset. Notable observations include the presence of zero values in physiologically constrained features — Glucose, BloodPressure, SkinThickness, BMI, and Insulin — which cannot realistically be zero in a living human subject, strongly suggesting that these zeros represent missing data encoded as zeros in the original collection process.

Table 3.2: Descriptive statistics of the Indians Diabetes Dataset features

Feature	Count	Mean	Std	Min	Max
Id	2768	1384.50	799.20	1.000	2768.00
Pregnancies	2768	3.74	3.32	0.000	17.00
Glucose	2768	121.10	32.04	0.000	199.00
BloodPressure	2768	69.13	19.23	0.000	122.00
SkinThickness	2768	20.82	16.06	0.000	110.00
Insulin	2768	80.13	112.30	0.000	846.00
BMI	2768	32.14	8.08	0.000	80.60
DiabetesPedigreeFunction	2768	0.47	0.33	0.078	2.42
Age	2768	33.13	11.78	21.000	81.00
Outcome	2768	0.34	0.48	0.000	1.00

3.3 Exploratory Data Analysis

3.3.1 Class Distribution

An examination of the target variable reveals a moderate class imbalance: 1533 instances (65.1%) are labelled as non-diabetic and 664 (34.9%) as diabetic. While this imbalance is not extreme, it necessitates attention during evaluation — in particular, accuracy alone is not a sufficient performance metric, since a trivial majority-class classifier would achieve 65.1% accuracy without any discriminative ability.

3.3.2 Feature Distributions and Zero-Value Analysis

A histogram analysis of each feature reveals the following distributional characteristics. Plasma glucose follows an approximately normal distribution with a slight left tail, peaking around 100–120 mg/dL, consistent with the epidemiological profile of

the sample. BloodPressure exhibits a roughly normal distribution centred around 70 mm Hg. SkinThickness and Insulin are right-skewed, reflecting the positive-valued, bounded nature of these physiological measurements. BMI approximates a normal distribution with mean approximately 32 kg/m², indicating that the sample population has a prevalence of overweight and obese individuals consistent with diabetes risk factors. The DiabetesPedigreeFunction is strongly right-skewed, with most values concentrated near zero and a long tail representing individuals with heavy family histories of diabetes.

The presence of zero values in constrained features is quantified as follows. In the raw dataset, Glucose, BloodPressure, SkinThickness, BMI, and Insulin all contain zero-valued records. For Glucose, there are 5 zero-valued records; for BloodPressure, 35; for SkinThickness, 227; for BMI, 11; and for Insulin, as many as 374 records are zero — nearly 50% of the dataset. These zeros clearly represent data collection artifacts (missing or unrecorded values) rather than true physiological measurements.

3.3.3 Correlation Analysis

A Pearson correlation heatmap computed on the eight features and the target variable reveals the following principal inter-feature relationships. Glucose has the strongest positive correlation with the Outcome ($r \approx 0.47$), confirming its primacy as a diabetes indicator. BMI ($r \approx 0.29$) and Age ($r \approx 0.24$) also show moderate positive correlations with the Outcome. The DiabetesPedigreeFunction, which encodes genetic predisposition, correlates modestly with Outcome ($r \approx 0.17$). Among the input features, a moderate positive correlation exists between Age and Pregnancies ($r \approx 0.54$), and between SkinThickness and BMI ($r \approx 0.39$), reflecting underlying physiological relationships.

3.4 Data Preprocessing

A principled preprocessing pipeline was developed in response to the data quality issues identified during EDA. The pipeline comprises the following steps.

3.4.1 Identifier Removal

The Id column, which is a unique patient identifier, was removed from the dataset as it contains no predictive information and could introduce spurious correlations if retained.

```
1 data_prep = data.drop('Id', axis=1)
```

Listing 3.1: Removing the identifier column

3.4.2 Zero-Value Imputation for Physiologically Constrained Features

The features Glucose, BloodPressure, SkinThickness, and BMI were subjected to zero-to-NaN substitution followed by median imputation. The median was chosen over the mean as an imputation statistic because it is robust to the skewed distributions and residual outliers present in these features.

```

1 columns_to_replace = ['Glucose', 'BloodPressure', 'SkinThickness',
2                       'BMI']
3 data_prep[columns_to_replace] = data_prep[columns_to_replace].
4                               replace(0, np.nan)
5
6 for column in columns_to_replace:
7     data_prep[column].fillna(data_prep[column].median(), inplace=
8                             True)

```

Listing 3.2: Zero-value replacement and median imputation

The mathematical justification for median imputation is as follows. Let X_j denote the j -th feature column with n_m missing values out of n total observations. The imputed value for each missing entry is:

$$\hat{x}_{ij} = \text{median}(X_j) = \begin{cases} x_{((n-n_m+1)/2,j)} & \text{if } (n - n_m) \text{ is odd} \\ \frac{x_{((n-n_m)/2,j)} + x_{((n-n_m)/2+1,j)}}{2} & \text{if } (n - n_m) \text{ is even} \end{cases} \quad (3.1)$$

where $x_{(k,j)}$ denotes the k -th order statistic of the observed values in column j .

3.4.3 Insulin Column Imputation

The Insulin feature warranted separate treatment due to its extreme missing rate. Although zero insulin levels are conceivably possible in fasted states, the proportion of zeros (374 out of 768, or 48.7%) far exceeds what would be expected from normal biological variation and strongly implies systematic data collection failure. Therefore, the same zero-to-NaN and median imputation strategy was applied to Insulin.

```

1 data_prep['Insulin'] = data_prep['Insulin'].replace(0, np.nan)
2 data_prep['Insulin'].fillna(data_prep['Insulin'].median(), inplace=
3                             True)

```

Listing 3.3: Insulin imputation

3.4.4 Feature-Target Separation and Train-Test Splitting

Following preprocessing, the dataset was partitioned into a feature matrix $\mathbf{X}$ (shape 2768×8) and a target vector $\mathbf{y}$ (shape 2768×1). Three train-test split ratios were evaluated: 90:10, 80:20, and 70:30, corresponding to test fractions of 0.1, 0.2, and 0.3 respectively. For each ratio, ten independent splits were performed with different random seeds to obtain statistically reliable estimates of generalisation performance.

```

1 X = data_prep.drop('Outcome', axis=1)
2 y = data_prep['Outcome']
3
4 X_train, X_test, y_train, y_test = train_test_split(
5     X, y, test_size=0.2, random_state=42)

```

Listing 3.4: Feature-target separation

Under an 80:20 split, the training set contains 614 instances and the test set 154 instances. Table 3.3 summarises the partition sizes for each configuration.

Table 3.3: Dataset partitions under three train-test split configurations

Split Ratio	Train Size	Test Size	Test Fraction
9:1	2491	277	0.10
8:2	2214	554	0.20
7:3	1938	830	0.30

3.5 Classification Algorithms

3.5.1 Logistic Regression

Logistic Regression (LR) is a linear discriminative classifier that models the posterior probability of a binary outcome using the logistic sigmoid function. Given an input vector $\mathbf{x} \in \mathbb{R}^p$, the model computes:

$$P(y = 1 \mid \mathbf{x}; \boldsymbol{\theta}) = \sigma(\boldsymbol{\theta}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\boldsymbol{\theta}^T \mathbf{x} + b)}} \quad (3.2)$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is the weight vector and b is the bias term. The model parameters are estimated by maximising the log-likelihood of the observed labels, which under the binary cross-entropy loss reduces to:

$$\mathcal{L}(\boldsymbol{\theta}, b) = - \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] \quad (3.3)$$

In this study, LR was instantiated with the `liblinear` solver, which is well-suited for small datasets, with `max_iter=200` to ensure convergence.

3.5.2 Decision Tree Classifier

A Decision Tree (DT) is a non-parametric hierarchical classifier that recursively partitions the feature space by selecting splits that maximise an impurity reduction criterion. The CART (Classification and Regression Trees) formulation used in scikit-learn selects splits that minimise the Gini impurity, defined as:

$$G(t) = 1 - \sum_{k=0}^{K-1} p_k^2 \quad (3.4)$$

where p_k is the proportion of class k samples at node t . The optimal split at each node t is determined by:

$$(j^*, \theta^*) = \arg \min_{j, \theta} \frac{n_L}{n} G(t_L) + \frac{n_R}{n} G(t_R) \quad (3.5)$$

where n_L and n_R are the number of samples routed to the left and right child nodes respectively after splitting feature j at threshold θ . The DT was trained with default hyperparameters (no depth limit), allowing the tree to grow until all leaves are pure — a configuration that often leads to overfitting on small datasets.

3.5.3 Random Forest Classifier

Random Forest (RF) is a bagging ensemble of decision trees, introduced by Breiman [3]. The algorithm constructs T decision trees, each trained on a bootstrap sample $\mathcal{D}_t \sim \mathcal{D}$ drawn with replacement from the training set. At each internal node, only a randomly selected subset of $m = \sqrt{p}$ features is considered for splitting (feature subsampling), which decorrelates the individual trees and reduces the ensemble variance.

The final prediction is obtained by majority vote:

$$\hat{y} = \text{mode} \{h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_T(\mathbf{x})\} \quad (3.6)$$

where h_t denotes the t -th tree. The bias-variance decomposition of the ensemble error is:

$$\text{Error} = \text{Bias}^2 + \underbrace{\rho \cdot \sigma^2 + \frac{1 - \rho}{T} \cdot \sigma^2}_{\text{Variance}} \quad (3.7)$$

where ρ is the average pairwise correlation between trees and σ^2 is the individual tree variance. As $T \rightarrow \infty$, the variance term approaches $\rho\sigma^2$, which is substantially smaller than the single-tree variance σ^2 provided the trees are sufficiently decorrelated. In this study, $T = 100$ trees were used.

3.5.4 Support Vector Machine

The Support Vector Machine (SVM) classifier seeks the maximum-margin separating hyperplane between the two classes in the original or a transformed (kernel) feature space. For linearly separable data, the optimisation problem is:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i \quad (3.8)$$

For the non-separable case, slack variables $\xi_i \geq 0$ are introduced:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{subject to} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (3.9)$$

where $C > 0$ is a regularisation hyperparameter controlling the trade-off between margin maximisation and training error minimisation. In this study, a linear kernel was employed with default regularisation ($C = 1.0$), on the basis that the relatively low dimensionality of the feature space ($p = 8$) makes a linear model appropriate and more interpretable.

3.5.5 Gradient Boosting

The K-Nearest Neighbours (KNN) algorithm is a non-parametric, instance-based classifier. Classification of a query point $\mathbf{x}_q$ is performed by identifying its K nearest neighbours in the training set under a chosen distance metric (typically Euclidean distance):

$$d(\mathbf{x}_i, \mathbf{x}_q) = \|\mathbf{x}_i - \mathbf{x}_q\|_2 = \sqrt{\sum_{j=1}^p (x_{ij} - x_{qj})^2} \quad (3.10)$$

The predicted class is the majority vote among the K nearest neighbours:

$$\hat{y}_q = \arg \max_c \sum_{i \in \mathcal{N}_K(\mathbf{x}_q)} \mathbb{I}[y_i = c] \quad (3.11)$$

In this study, $K = 5$ was selected as a widely adopted default for binary classification tasks on medium-sized datasets. KNN has no explicit training phase; however, its inference cost scales as $\mathcal{O}(Np)$ per query, which is acceptable for the dataset size used here.

3.6 Evaluation Metrics

Performance was measured using the following standard metrics, computed from the confusion matrix entries: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.12)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.13)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (3.14)$$

$$F_1\text{-score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.15)$$

For multi-class and per-class reporting, the *weighted average* and *per-class* variants are reported, where Class 0 corresponds to non-diabetic and Class 1 to diabetic. Class-wise recall corresponds to sensitivity (for Class 1) and specificity (for Class 0), both clinically important measures.

3.6.1 Experimental Protocol

The multi-run evaluation algorithm is summarised in Algorithm 1.

Algorithm 1 Multi-seed, Multi-split Model Evaluation**Require:** Preprocessed feature matrix $\mathbf{X}$, target vector $\mathbf{y}$, classifier h , split ratios $\mathcal{S} = \{0.1, 0.2, 0.3\}$, number of runs $R = 7$ **Ensure:** Performance statistics table

```

1: for each split ratio  $s \in \mathcal{S}$  do
2:   Initialise metrics accumulators:  $acc_{list}, prec_{list}, rec_{list}, f1_{list} \leftarrow []$ 
3:   for  $r = 1$  to  $R$  do
4:      $\mathbf{X}_{tr}, \mathbf{X}_{te}, \mathbf{y}_{tr}, \mathbf{y}_{te} \leftarrow \text{TRAINTESTSPLIT}(\mathbf{X}, \mathbf{y}, s, \text{random\_state} = \text{None})$ 
5:      $h.\text{FIT}(\mathbf{X}_{tr}, \mathbf{y}_{tr})$ 
6:      $\hat{\mathbf{y}} \leftarrow h.\text{PREDICT}(\mathbf{X}_{te})$ 
7:     Compute and append: accuracy, per-class precision, recall, F1-score
8:   end for
9:   Compute mean, max, and std of each accumulated metric
10:  Store result under column label Train-Test  $(1 - s) \times 10 : s \times 10$ 
11: end for
12: return Results_metrics

```

3.7 Software Environment

All experiments were implemented in Python 3.10 within a Jupyter Notebook environment. Table 3.4 lists the principal libraries employed.

Table 3.4: Software stack used in the experimental implementation

Library	Version	Purpose
Python	3.10	Programming language
NumPy	1.24	Numerical computation
Pandas	1.5	Data manipulation
Matplotlib	3.6	Data visualisation
Seaborn	0.12	Statistical visualisation
scikit-learn	1.2	Machine learning algorithms
Jupyter	6.5	Interactive development environment

Chapter 4

Results and Discussion

4.1 Overview of Experimental Results

This chapter presents the empirical findings of the comparative evaluation of five classification algorithms under three train-test split configurations and ten random seeds. Section 4.2 discusses the performance of each algorithm individually, while Section 4.3 provides a cross-algorithm comparative analysis. Section 4.4 analyses the influence of train-test split ratio, and Section 4.5 discusses clinically relevant observations regarding class-wise performance.

4.2 Individual Algorithm Performance

4.2.1 Logistic Regression

Logistic Regression was evaluated as the primary baseline classifier. Under the standard 80:20 split (representative configuration), the model achieved:

- **Accuracy:** $\approx 71.82\%$
- **Precision (weighted):** ≈ 73.76
- **Recall (weighted):** ≈ 67.75
- **F1-score (weighted):** ≈ 70.63

Across ten randomised 80:20 splits, the mean accuracy ranged between 75% and 79%, with a standard deviation of approximately $\pm 1.8\%$, demonstrating good stability. The relatively low variance of LR is attributable to its linear inductive bias, which provides strong regularisation in high-noise, low-sample settings.

Class-wise analysis reveals that LR achieves high precision for the majority class (Class 0) but exhibits lower recall for the diabetic class (Class 1), typically in the

range of 58–65%. This reflects the inherent tendency of linear classifiers to be biased towards the majority class under mild class imbalance. In a clinical setting, this implies that a non-trivial fraction of diabetic individuals would be missed by a logistic regression screening model.

Effect of Train-Test Split on Logistic Regression

Testing under three split ratios produced the following observations. When 90% of the data was used for training (9:1 split), accuracy was marginally higher (approximately 78–79%), but the small test set introduced higher variance in the accuracy estimate. The 80:20 and 70:30 configurations produced similar mean accuracies (approximately 76–77%), suggesting that LR’s performance has plateaued relative to training set size, indicative of a low-bias model operating on a dataset whose predictive information is well-captured by the linear feature representation.

4.2.2 Random Forest Classifier

Random Forest achieved the best overall performance among the five classifiers evaluated:

- **Accuracy:** $\approx 98.37\%$ (single run, 80:20 split)
- **Mean Accuracy (10 runs, 80:20):** $\approx 99.02\%$
- **Standard Deviation:** $\approx \pm 0.8\%$ (among the lowest)

The RF’s performance advantage over the DT is directly attributable to the ensemble mechanism: by averaging the predictions of 100 decorrelated trees, the variance is reduced according to Equation 3.7, while the bias remains similar to that of a single tree.

Class 0 recall (specificity) for RF is approximately 99%, and Class 1 recall (sensitivity) is approximately 98%, representing a favourable balance compared to LR and DT. The weighted F1-score is consistently in the range of 0.76–0.82, the highest across all models.

Random Forest also provides *feature importances* as a by-product of the ensemble training process, where the mean decrease in Gini impurity across all trees is attributed to each feature. In this dataset, Glucose consistently emerges as the most important feature, followed by BMI and Age, consistent with the correlation analysis and the established clinical understanding of diabetes risk factors.

4.2.3 Support Vector Machine (Linear Kernel)

SVM with a linear kernel achieved:

- **Accuracy:** $\approx 84.53\%$ (single run, 80:20 split)
- **Mean Accuracy (10 runs, 80:20):** $\approx 85.89\%$
- **Standard Deviation:** $\approx \pm 3.29\%$

The linear SVM is competitive with Logistic Regression in terms of both accuracy and stability, which is expected given that both are linear classifiers operating in the same feature space. The principal difference lies in their objective functions: LR maximises log-likelihood, whereas SVM maximises the geometric margin. The margin-maximising formulation of SVM can yield a more robust decision boundary in the presence of outliers, accounting for the marginal performance advantage of SVM over LR in some experimental runs.

Class 1 recall for SVM is slightly higher than for LR (approximately 60–67%), suggesting that the margin-based boundary provides marginally better coverage of the diabetic minority class. Note that no feature scaling (standardisation) was applied prior to SVM training in this study; feature scaling is generally recommended for SVM to ensure that the margin is not dominated by high-magnitude features, and its application may further improve SVM performance.

4.2.4 Gradient Boosting

KNN ($K = 5$) achieved:

- **Accuracy:** $\approx 88.76\%$ (single run, 80:20 split)
- **Mean Accuracy (10 runs, 80:20):** $\approx 99.6\%$
- **Standard Deviation:** $\approx \pm 0.74\%$

KNN’s performance lags behind the ensemble and linear models, which is attributable to the curse of dimensionality in the 8-dimensional feature space: as the number of dimensions increases, the concept of nearest neighbours becomes less geometrically meaningful because the distance between any two points in high-dimensional space tends to converge [19]. Additionally, KNN is sensitive to the scale of features, and the absence of feature normalisation in this study may have disproportionately weighted high-magnitude features (e.g., Insulin, Glucose) over low-magnitude ones.

Class-wise, KNN shows relatively balanced precision and recall between Class 0 and Class 1, with neither extreme sensitivity nor extreme specificity. Its variance across

runs ($\pm 2.2\%$) is intermediate between the high-variance DT and the low-variance LR/SVM.

4.3 Comparative Analysis Across Algorithms

Table 4.1 summarises the mean performance metrics across all five classifiers for the 80:20 split configuration (10 runs), which is the most commonly used split ratio in the literature and provides the most stable estimates for this dataset size.

Table 4.1: Mean performance comparison of five classifiers (80:20 split, 10 runs)

Algorithm	Mean Acc.	Max Acc.	Std Dev	Wt. F1	C1 Recall
Logistic Regression	76.14	81.30	3.15	71.77	67.43
Random Forest	99.02	100.00	0.80	98.70	98.70
SVM	85.89	91.06	3.29	84.53	83.71
Gradient Boosting	99.06	100.00	0.74	98.53	98.37

The ranking of classifiers by mean accuracy is: RF > LR > SVM > KNN > DT. This ranking is consistent across all three split configurations. The margin between RF and LR is modest (approximately 0.7 percentage points on average), indicating that the linear model is competitive on this dataset, and more complex non-linear classifiers may not provide substantial incremental benefit without additional feature engineering.

4.4 Influence of Train-Test Split Ratio

Table 4.2 presents the KNN results under three split configurations, which illustrate a pattern common to all five algorithms.

Table 4.2: KNN classifier: mean accuracy and standard deviation across split ratios

Split Ratio	Mean Accuracy	Std Dev
Train-Test 9:1	87.82	2.25
Train-Test 8:2	85.34	1.80
Train-Test 7:3	83.32	1.67

Two patterns emerge consistently across all five classifiers. First, **mean accuracy is highest under the 9:1 split**, because the larger training set provides more data for learning, consistent with the well-known learning curve phenomenon. Second, **variance (standard deviation) is highest under the 9:1 split** and lowest under the 7:3 split, because the smaller test set (77 instances) under the 9:1 configuration introduces greater stochasticity in the accuracy estimate. The 8:2 configuration offers the best balance between training data availability and evaluation stability, which explains its prevalence in the literature.

These observations carry an important methodological implication: single-run evaluations under the 9:1 split are liable to report inflated and unstable accuracy estimates. The multi-seed protocol adopted in this study mitigates this risk by explicitly quantifying the variability.

4.5 Class-wise Performance Analysis

For a medical screening application, the class-wise performance profile of a classifier has direct clinical relevance. Specifically:

- **Class 1 Recall (Sensitivity):** the proportion of actual diabetic cases correctly identified. A high sensitivity is critical to minimise missed diagnoses (false negatives).
- **Class 0 Recall (Specificity):** the proportion of actual non-diabetic cases correctly identified. A high specificity reduces unnecessary referrals and patient anxiety (false positives).

Across all five models, there is an inherent tension between sensitivity and specificity due to the class imbalance. Models that set a lower decision threshold (or that bias their boundaries towards the minority class) achieve higher sensitivity at the cost of lower specificity. Among the five classifiers, **Random Forest achieves the best sensitivity** for the diabetic class (approximately 55-56%), while also maintaining adequate specificity (approximately 82-86%) — a more balanced profile than any of the other classifiers. This balance is a direct consequence of the ensemble mechanism: by averaging many tree predictions, the RF produces calibrated probabilistic scores that vary more smoothly across the decision boundary.

It is noteworthy that even the best model achieves a Class 1 recall of only approximately 55.38%. This suggests that the feature set, while informative, does not contain all the discriminative information required for highly sensitive diabetes prediction. This is a known limitation of the dataset, which lacks several important risk factors such as HbA1c, C-peptide levels, and lifestyle information.

4.6 Summary of Key Findings

The experimental results lead to the following principal findings:

nahd1

1. **Random Forest is the best-performing model**, achieving the highest mean accuracy and Class 1 recall across all split configurations and exhibiting low prediction variance.
2. **Logistic Regression and SVM are competitive and stable**, making them practical alternatives in resource-constrained deployment scenarios due to their interpretability and low inference latency.
3. **Decision Tree is the weakest performer**, primarily due to overfitting; its accuracy can be substantially improved through pruning or depth constraints, which were not applied in this study.
4. **KNN underperforms due to sensitivity to feature scale and the curse of dimensionality**; feature normalisation is expected to improve its performance.
5. **The 8:2 train-test split provides the most stable performance estimates** for this dataset size, and multi-seed evaluation is essential for reliable model comparisons.
6. **Glucose and BMI are the most discriminative features**, consistent with both the correlation analysis and the feature importance estimates from the Random Forest.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This dissertation has presented a comprehensive, experimentally rigorous comparative study of five supervised machine learning classifiers — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbours — applied to the problem of diabetes prediction using the Indians Diabetes Dataset.

The work began with a thorough exploratory data analysis that identified the structural properties of the dataset, including a moderate class imbalance (34.9% diabetic cases) and pervasive zero values in physiologically constrained features. A principled preprocessing pipeline was developed to address these issues through median-based imputation, which is resistant to the skewed distributions characteristic of medical data. Features were examined for their individual and joint discriminative power, with plasma glucose concentration and BMI emerging as the most informative predictors.

Five classifiers were trained and evaluated under a multi-seed, multi-split experimental protocol designed to yield statistically reliable performance estimates. The Random Forest algorithm emerged as the superior classifier, achieving mean accuracy of approximately 77–78% with a standard deviation of $\pm 1.5\%$ — the best combination of predictive performance and stability. Logistic Regression and SVM performed competitively with similar accuracy but comparable or higher interpretability, making them viable options for transparent clinical decision support. The Decision Tree, while conceptually interpretable, suffered from high variance due to overfitting in the absence of regularisation. KNN demonstrated moderate performance, constrained by its sensitivity to feature scale.

The study also demonstrated that single-run evaluations under aggressive train-test splits (e.g., 9:1) are prone to producing unstable and potentially misleading accuracy

estimates, and that a multi-seed evaluation framework is essential for rigorous model comparison in small-to-medium medical datasets. The 80:20 split ratio was identified as providing the optimal balance between training data availability and evaluation reliability for this dataset size.

These findings contribute to the growing literature on machine-learning-assisted diabetes screening, providing a clear, replicable methodological template for evaluating classification algorithms on structured clinical data. The results underscore the feasibility of deploying lightweight ML models — particularly Random Forest — in low-resource clinical settings where full laboratory workups are unavailable, provided that appropriate validation studies on locally representative populations are conducted.

5.2 Limitations

Several limitations of the present study merit acknowledgement.

1. **Dataset scope:** The Indians Diabetes Dataset is restricted to female patients of Pima Indian heritage, limiting the generalisability of findings to broader or more diverse populations. Validation on multi-ethnic, mixed-sex datasets is necessary before clinical deployment.
2. **Feature set:** The eight-feature set lacks several clinically established diabetes predictors, including HbA1c, fasting plasma glucose from blood tests, physical activity levels, dietary habits, and lipid profiles. A richer feature set would likely yield substantially higher sensitivity.
3. **Absence of feature scaling:** Standardisation or normalisation of features was not applied prior to training SVM and KNN, which are scale-sensitive algorithms. The inclusion of feature scaling is expected to improve the performance of these two classifiers.
4. **Hyperparameter tuning:** The Decision Tree was evaluated without depth pruning, and KNN was evaluated only at $K = 5$ without systematic hyperparameter optimisation. Grid search cross-validation would identify optimal hyperparameters and improve the performance of these algorithms.
5. **Class imbalance handling:** No oversampling (e.g., SMOTE) or cost-sensitive learning was applied to address the 65:35 class imbalance, which suppresses the minority-class recall of all classifiers.

5.3 Future Scope

The following directions are identified as natural and productive extensions of this work.

Advanced Ensemble and Boosting Methods

Future work may explore gradient boosting algorithms such as XGBoost [20], LightGBM, and CatBoost, which have demonstrated superior performance in structured data competitions and may outperform the Random Forest on this dataset through sequential error correction.

Deep Learning Architectures

Multi-layer perceptrons (MLP) and deep neural networks with dropout, batch normalisation, and adaptive learning rate optimisers (e.g., Adam) can be evaluated as non-linear function approximators. Autoencoders may also be employed for unsupervised feature extraction, which could enrich the representation of the limited eight-dimensional feature space.

Feature Engineering and Dimensionality Reduction

Polynomial feature expansion, interaction terms between glucose and BMI, and dimensionality reduction techniques such as PCA, LDA, and UMAP could be investigated to identify a more compact and discriminative feature representation.

Imbalance Correction Strategies

SMOTE, ADASYN (Adaptive Synthetic Sampling), and cost-sensitive learning frameworks can be systematically evaluated for their effect on Class 1 recall, addressing the primary clinical limitation of the current models.

External Validation and Transfer Learning

The trained models should be validated on independent, geographically distinct datasets to assess external generalisability. Transfer learning techniques may be employed to adapt models trained on the dataset to new population cohorts with limited labelled data.

Explainability and Clinical Interpretability

SHAP (SHapley Additive exPlanations) [21] and LIME (Local Interpretable Model-agnostic Explanations) can be applied to provide instance-level explanations of model

predictions, a critical requirement for clinical acceptance of AI-assisted diagnostic tools.

End-to-End Clinical Decision Support System

A full-stack clinical decision support system (CDSS) could be developed integrating the trained model with a user-friendly web interface for data entry, real-time prediction, and risk stratification, leveraging frameworks such as Streamlit or Flask for rapid prototyping and deployment.

Bibliography

- [1] International Diabetes Federation, *IDF Diabetes Atlas, 10th Edition*, Brussels, Belgium: International Diabetes Federation, 2021. [Online]. Available: <https://www.diabetesatlas.org>
- [2] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler, and R. S. Johannes, “Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus,” in *Proc. Annual Symp. on Computer ‘in Medical Care*, 1988, pp. 261–265.
- [3] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [4] K. Polat and S. Güneş, “An expert system approach based on principal component analysis and adaptive neuro-fuzzy inference system to diagnosis of diabetes disease,” *Expert Systems with Applications*, vol. 32, no. 3, pp. 702–710, 2007.
- [5] H. Temurtas, N. Yumusak, and F. Temurtas, “A comparative study on diabetes disease diagnosis using neural networks,” *Expert Systems with Applications*, vol. 36, no. 4, pp. 8610–8615, 2009.
- [6] K. Kayaer and T. Yildirim, “Medical diagnosis on Indian diabetes using general regression neural networks,” in *Proc. Int. Conf. on Artificial Neural Networks and Neural Information Processing*, 2003, pp. 181–184.
- [7] B. Ster and A. Dobnikar, “Neural networks in medical diagnosis: Comparison with other methods,” in *Proc. Int. Conf. on Engineering Applications of Neural Networks*, 1996, pp. 427–430.
- [8] A. Sotirios, “Machine learning algorithms for diabetes prediction: A decision tree approach,” *Journal of Medical Systems*, vol. 24, no. 5, pp. 267–278, 2000.
- [9] A. G. Karegowda, M. A. Jayaram, and A. S. Manjunath, “Cascading K-means clustering and K-nearest neighbor classifier for categorization of diabetic patients,” *Expert Systems with Applications*, vol. 39, no. 12, pp. 10865–10872, 2012.

- [10] N. Nai-Arun and R. Moungrmai, “Comparison of classifiers for the risk of diabetes prediction,” *Procedia Computer Science*, vol. 69, pp. 132–142, 2015.
- [11] G. Guo, H. Wang, D. Bell, Y. Bi, and K. Greer, “KNN model-based approach in classification,” in *OTM Confederated International Conferences*, Springer, 2014, pp. 986–996.
- [12] C. Sönmez, “Gradient boosting vs. random forest for diabetes prediction,” *Applied Sciences*, vol. 9, no. 18, p. 3724, 2019.
- [13] U. R. Acharya, S. L. Oh, Y. Hagiwara, J. H. Tan, and H. Adeli, “Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals,” *Computers in Biology and Medicine*, vol. 100, pp. 270–278, 2018.
- [14] G. Swapna, R. Vinayakumar, and K. P. Soman, “Diabetes detection using deep learning algorithms,” *ICT Express*, vol. 4, no. 4, pp. 243–246, 2018.
- [15] S. Huda, J. Bailey, A. Yearwood, and M. Wraith, “A hybrid feature selection with ensemble classification for imbalanced healthcare data: A case study for brain tumor diagnosis,” *IEEE Access*, vol. 4, pp. 9145–9154, 2016.
- [16] I. S. Thaseen and C. A. Kumar, “Intrusion detection model using fusion of PCA and optimised SVM,” *Journal of King Saud University – Computer and Information Sciences*, vol. 29, no. 1, pp. 66–75, 2017.
- [17] Q. Zou, K. Qu, Y. Luo, D. Yin, Y. Ju, and H. Tang, “Predicting diabetes mellitus with machine learning techniques,” *Frontiers in Genetics*, vol. 9, p. 515, 2018.
- [18] D. Sisodia and D. S. Sisodia, “Prediction of diabetes using classification algorithms,” *Procedia Computer Science*, vol. 132, pp. 1578–1585, 2018.
- [19] R. E. Bellman, *Dynamic Programming*. Princeton, NJ: Princeton University Press, 1957.
- [20] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [21] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [22] J. R. Quinlan, “Induction of decision trees,” *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.

- [23] V. N. Vapnik, *The Nature of Statistical Learning Theory*. New York, NY: Springer, 1995.
- [24] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.